From Commutation Relations to Growth Diagrams: A Summary of the Operator–Bijection Correspondence

Summary of work by Robin Langer

Abstract

This note summarizes the relationship between three pieces of work: (1) the theory of row and column operators on symmetric functions developed in Part 1 of *Symmetric Functions and Macdonald Polynomials* (Masters thesis, 2009), (2) the (q,t)-vertex operator commutation relations used in *Enumeration of Cylindric Plane Partitions – Part I* (2012), and (3) the cylindric growth diagrams providing a bijective proof of Borodin's identity (PhD thesis). The central theme is that each algebraic commutation relation between row/column operators has a precise bijective counterpart in the form of a local rule applied within a growth diagram.

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1 The ring of symmetric functions and its operators

1.1 Setup and notation

Let Λ denote the ring of symmetric functions with natural basis $\{s_\lambda\}$ (the Schur functions), and let Λ^* denote the dual space with respect to the Hall inner product. We use the *plethystic notation* of Garsia–Haiman–Tesler: writing $X = x_1 + x_2 + \cdots$ for a set of indeterminates, the complete and elementary symmetric function generating series are

$$\Omega(Xz) = \Omega_z(X) = \prod_{x \in X} \frac{1}{1 - xz} = \sum_{n \geq 0} h_n(X) z^n,$$

$$\tilde{\Omega}(Xz) = \tilde{\Omega}_z(X) = \prod_{x \in X} (1 + xz) = \sum_{n \geq 0} e_n(X) z^n.$$

The *Hall inner product* satisfies $\langle m_\lambda(Y), h_\mu(X) \rangle = \delta_{\lambda\mu}$, or equivalently

$$\sum_\lambda P_\lambda(X) Q_\lambda(Y) = \Omega(XY) = \prod_{\substack{x \in X \\ y \in Y}} \frac{1}{1 - xy}$$

for any pair of dual bases $\{P_\lambda\}$, $\{Q_\lambda\}$.

1.2 Translation and multiplication operators

Following Garsia–Haiman–Tesler, define the *translation* and *multiplication* operators:

$$\mathcal{T}(z)[f(X)] = f(X + z), \tag{1}$$

$$\mathcal{P}(z)[f(X)] = \Omega(Xz) \cdot f(X). \tag{2}$$

In terms of the dual bases $\{m_\lambda(Y)\}$ and $\{h_\lambda(X)\}$:

$$\mathcal{T}(z) = \sum_n z^n \partial_{h_n}, \qquad \mathcal{P}(z) = \sum_n z^n h_n(X),$$

and dually $\mathcal{T}(z) = \mathcal{P}^*(z)$ (the adjoint with respect to the Hall inner product).

1.3 The fundamental commutation relation

The following identity governs how translation and multiplication interact:

Lemma 1 (Masters thesis, Lemma 1.1)**.**

$$\mathcal{T}(v)\,\mathcal{P}(w) = \Omega(vw)\,\mathcal{P}(w)\,\mathcal{T}(v). \tag{3}$$

Proof. $\mathcal{T}(v)\mathcal{P}(w)[h(X)] = \mathcal{T}(v)[\Omega(Xw)h(X)] = \Omega((X+v)w)h(X+v) = \Omega(vw)\Omega(Xw)\mathcal{T}(v)[h(X)]$. □

This single identity is the algebraic engine behind all of the enumerative results. The scalar factor $\Omega(vw) = 1/(1-vw)$ (when v, w are single variables) is precisely the generating function $\sum_{n\geq 0}(vw)^n$ which counts the "extra" horizontal strips.

1.4 Row and column operators

The *column operator* and *row operator* are defined as:

$$C(z) = \sum_m (-1)^m C_m z^m = \mathcal{P}(-z)\,\mathcal{T}(1/z), \tag{4}$$

$$R(z) = \sum_m z^m (-1)^m R_m = \mathcal{P}(z)\,\mathcal{T}(-1/z) = C(-z). \tag{5}$$

Proposition 2 (Masters thesis, Prop. 1.4)**.** *The Schur function has a natural expression as:*

$$s_\lambda(X) = R_{\lambda_1} \circ R_{\lambda_2} \circ \cdots \circ R_{\lambda_n}[1] = C_{\lambda'_1} \circ C_{\lambda'_2} \circ \cdots \circ C_{\lambda'_m}[1].$$

This is the operator-theoretic avatar of the Jacobi–Trudi and Nägelsbach–Kostka determinantal formulas.

1.5 Commutation of column operators with translation

From (**??**) one derives:

$$\mathcal{T}(v)\,C(z) = \mathcal{T}(v)\,\mathcal{P}(-z)\,\mathcal{T}(1/z) = \Omega(-vz)\,\mathcal{P}(-z)\,\mathcal{T}(v)\,\mathcal{T}(1/z) = (1-vz)\,C(z)\,\mathcal{T}(v), \tag{6}$$

which yields the commutator identity:

$$[C(z), \mathcal{T}(v)] = vz\,C(z)\,\mathcal{T}(v). \tag{7}$$

Extracting the coefficient of $(-1)^m v^k z^m$:

$$[C_m, \partial_{h_k}] = -C_{m-1}\partial_{h_{k-1}},$$

or equivalently $\partial_{h_m} C_k = C_k \partial_{h_m} + C_{k-1}\partial_{h_{m-1}}$.

Pieri rules from commutation. Applying $\mathcal{T}(z)$ to $s_\lambda(X) = C_{\lambda'_1} \circ \cdots \circ C_{\lambda'_m}[1]$ and commuting $\mathcal{T}(z)$ through each column operator via (**??**) produces:

$$\mathcal{T}(z)[s_\lambda(X)] = s_\lambda(X+z) = \sum_{\mu \in D(\lambda)} s_\mu(X)\, z^{|\lambda|-|\mu|}, \tag{8}$$

where $D(\lambda)$ is the set of partitions obtained from λ by removing a horizontal strip. This is the **first form of the Pieri rule**.

Dually, $\mathcal{P}(z)[s_\mu(Y)] = s_\mu(Y)\Omega_z(Y) = \sum_{\lambda \in U(\mu)} s_\lambda(Y) z^{|\lambda|-|\mu|}$ gives the second form: the **Pieri formula** for adding a horizontal strip.

2 Generalized operators and Littlewood–Richardson bases

2.1 Umbral operators

Given an invertible formal power series $f(z)$ with compositional inverse $g(z)$, define the *umbral operator* U_f by:

$$U_f[\Omega(Xz)] = \Omega_f(Xz) = \prod_{x \in X} \frac{1}{1 - f(z)x} = \sum_n r_n(X) z^n,$$

extended multiplicatively. Then $U_f^{-1} = U_g$.

2.2 Generalized column and row operators

The *generalized column operator* is:

$$C_f(z) = U_f \circ C(z) \circ U_g = \mathcal{P}_{-f}(z) \circ \mathcal{T}_{1/g}(z),$$

and the *generalized row operator* $R_f(z) = C_f(-z)$.

Proposition 3 (Masters thesis, Prop. 1.8)**.** *The basis defined by* $P_\lambda(X) = C^f_{\lambda'_1} \circ C^f_{\lambda'_2} \circ \cdots \circ C^f_{\lambda'_m}[1]$ *is of Littlewood–Richardson type:* $P_\lambda(X + X') = \sum_{\mu,\nu} c^\lambda_{\mu\nu} P_\mu(X) P_\nu(X')$.

The key point: **the same commutation relation** (**??**), conjugated by U_f, governs all generalized LR bases. The generalized Pieri rules follow identically:

$$\mathcal{T}(z)[P_\lambda(X)] = \sum_{\mu \in D(\lambda)} P_\mu(X) \cdot P_{(|\lambda|-|\mu|)}(z).$$

3 Lifting to Macdonald polynomials and (q,t)-operators

3.1 The (q,t)-deformed inner product

In the Macdonald setting, the ring $\Lambda_{q,t}$ is equipped with the deformed inner product associated to the (q,t)-*Cauchy kernel*:

$$\Omega_{q,t}[XY] = \Omega\left[XY\frac{1-t}{1-q}\right] = \prod_{i,j} \frac{(tx_iy_j;q)_\infty}{(x_iy_j;q)_\infty}. \tag{9}$$

The Macdonald polynomials $\{P_\lambda(X;q,t)\}$ are orthogonal with respect to this inner product, with norms $b_\lambda(q,t) = \prod_{s \in \lambda} \frac{1-q^{a(s)+1}t^{\ell(s)}}{1-q^{a(s)}t^{\ell(s)+1}}$.

3.2 (q,t)-Pieri formulas

The Pieri rules for Macdonald polynomials take the form (Macdonald, Ch. VI):

$$\Omega_{q,t}[Xz]\,P_\mu(X;q,t) = \sum_{\lambda\in U(\mu)} \psi_{\lambda/\mu}(q,t)\,P_\lambda(X;q,t)\,z^{|\lambda|-|\mu|}, \tag{10}$$

$$\Omega^*_{q,t}[Xz]\,P_\lambda(X;q,t) = \sum_{\mu\in D(\lambda)} \varphi_{\lambda/\mu}(q,t)\,P_\mu(X;q,t)\,z^{|\lambda|-|\mu|}, \tag{11}$$

where $\psi_{\lambda/\mu}$ and $\varphi_{\lambda/\mu}$ are explicit products over arm-lengths and leg-lengths of boxes in λ and μ.

3.3 The (q,t)-commutation relation

The following is the (q,t)-analog of the Jimbo–Miwa commutation relation for vertex operators, due to Garsia–Haiman–Tesler:

Lemma 4 (CPP Part I, Lemma 4.2)**.**

$$\Omega^*_{q,t}[Xu]\,\Omega_{q,t}[Xv] = \frac{(tuv;q)_\infty}{(quv;q)_\infty}\,\Omega_{q,t}[Xv]\,\Omega^*_{q,t}[Xu]. \tag{12}$$

When $q=t$ (the Schur/Hall–Littlewood specialization), this reduces to

$$\Omega^*[Xu]\,\Omega[Xv] = \frac{1}{1-uv}\,\Omega[Xv]\,\Omega^*[Xu],$$

which is precisely the commutation $\mathcal{T}(v)\mathcal{P}(w) = \Omega(vw)\mathcal{P}(w)\mathcal{T}(v)$ from (**??**), since $\Omega^*[Xz] = \mathcal{T}(-z)$ (the adjoint/translation operator) when specialized appropriately.

3.4 The degree operator

In enumerating cylindric plane partitions, we need an additional operator D_z which tracks the total weight:

$$D_z\,P_\lambda[X] = z^{|\lambda|}\,P_\lambda[X].$$

Its commutation with the vertex operators is:

$$D_z\,\Omega_{q,t}[Xu] = \Omega_{q,t}[Xuz]\,D_z, \tag{13}$$

$$D_z\,\Omega^*_{q,t}[Xu] = \Omega^*_{q,t}[Xuz^{-1}]\,D_z. \tag{14}$$

4 Cylindric plane partitions as operator traces

4.1 From interlacing sequences to operator products

A cylindric plane partition of profile $\pi = (\pi_1,\dots,\pi_T)$ is a periodic interlacing sequence $(\mu^0,\mu^1,\dots,\mu^T)$ with $\mu^0=\mu^T$, where μ^k/μ^{k-1} is a horizontal strip if $\pi_k=1$, and μ^{k-1}/μ^k is a horizontal strip if $\pi_k=0$.

This structure means: **a CPP is built by successively applying $\Omega_{q,t}$ (adding strips) and $\Omega^*_{q,t}$ (removing strips)**. The generating function is a *trace*:

$$\sum_{\mathfrak{c}\in\mathrm{CPP}(\pi)} W_{\mathfrak{c}}(q,t)z^{|\mathfrak{c}|} = \sum_\mu \langle Q_\mu \mid G^{\pi_0}(u_0)\,G^{\pi_1}(u_1)\cdots G^{\pi_T}(u_T)\,D_w\,P_\mu\rangle_{q,t} \tag{15}$$

where $G^1(z)=\Omega_{q,t}[Xz]$, $G^0(z)=\Omega^*_{q,t}[Xz]$, and u_k, w are determined by the profile.

4.2 Proof of Borodin's identity via commutation

The proof of the (q,t)-Borodin formula (Theorem 1.1 of CPP Part I) proceeds by:

1. **Express** the CPP generating function as the trace (**??**).
2. **Commute** all $\Omega^*_{q,t}$ operators past all $\Omega_{q,t}$ operators using Lemma **??**. Each commutation produces a scalar factor $(tuv;q)_\infty/(quv;q)_\infty$.
3. **Shift** the D_w operator using (**??**)–(**??**) and iterate, producing:

$$M_\pi(m) = \prod_{\substack{(i,j)\\ \pi_i \neq \pi_j}} \frac{(tu_iu_jw^{m+1};q)_\infty}{(u_iu_jw^{m+1};q)_\infty}\, M_\pi(m+1).$$

4. **Take the limit** $m \to \infty$, where $M_\pi(\infty) = \prod_{n\geq 1} 1/(1-w^n)$.

Each commutation step produces one factor in the infinite product formula.

5 Growth diagrams: the bijective realization

5.1 The fundamental correspondence

The PhD thesis establishes a weight-preserving bijection:

$$\psi_\pi : (\mathrm{IP}, \mathrm{ALCD}(\pi)) \xrightarrow{\sim} \mathrm{CPP}(\pi), \tag{16}$$

where IP is the set of integer partitions and $\mathrm{ALCD}(\pi)$ the set of arbitrarily labelled cylindric diagrams with profile π. This bijection is encoded by a **cylindric growth diagram**.

5.2 Cylindric growth diagrams

Definition 5. For $\pi \in \mathrm{bin}(n,m)$, the *cylindric poset* $\mathfrak{G}(n,m)$ is the quotient of $\mathbb{Z}^2$ by $(x,y) \equiv (x+n, y-m)$. A *cylindric growth diagram* is a labelling of vertices by integer partitions and faces by non-negative integers, satisfying:

1. If $v \lhd w$ then $\lambda(v)/\lambda(w)$ is a horizontal strip.
2. All but finitely many vertices carry the same partition γ.
3. If $(u; v_1, v_2; w)$ is a face with label m, and $\mu = \lambda(u)$, $\alpha = \lambda(v_1)$, $\beta = \lambda(v_2)$, $\lambda = \lambda(w)$, then $\lambda = \mathfrak{U}_{\alpha,\beta}(m,\mu)$ where $\mathfrak{U}$ is the *local rule*.

The upper boundary path gives a cylindric plane partition; forgetting vertex labels gives an arbitrarily labelled cylindric diagram. The growth diagram *interpolates* between the two sides of the bijection.

5.3 Local rules as commutation relations

This is the key conceptual bridge.

The *local rule* $\mathfrak{U}_{\alpha,\beta}$ has type signature:

$$\mathfrak{U}_{\alpha,\beta} : U(\alpha) \cap U(\beta) \to (\mathbb{Z}_{\geq 0},\ D(\alpha) \cap D(\beta)),$$

where $U(\mu)$ (resp. $D(\lambda)$) is the set of partitions obtained by adding (resp. removing) a horizontal strip. It satisfies the weight condition: if $\lambda = \mathfrak{U}_{\alpha,\beta}(m, \mu)$ then

$$|\lambda| = |\alpha| + |\beta| - |\mu| + m. \tag{17}$$

The local rule is used as a *higher-order function*:

$$\mathfrak{L}_i : \big((\mathrm{IP}, \mathrm{ALCD}(\pi)) \to \mathrm{CPP}(\pi)\big) \ \longrightarrow\ \big((\mathrm{IP}, \mathrm{ALCD}(\pi')) \to \mathrm{CPP}(\pi')\big), \tag{18}$$

where $\pi \prec \pi'$ (obtained by adding one inversion).

Remark 6 (Algebra $\leftrightarrow$ Bijection dictionary)**.**

Algebraic proof	**Bijective proof**
Commuting $\Omega^*[Xu]$ past $\Omega[Xv]$	Applying local rule $\mathfrak{L}_i$
Scalar factor $\Omega(uv) = \frac{1}{1-uv}$	Face label m (non-negative integer)
Cyclic invariance of trace	Rotation of the cylinder
D_w shift (Lemma 5.1)	Depth increment in growth diagram
$M_\pi(m) \to M_\pi(m+1)$ iteration	Adding a new row of faces
$M_\pi(\infty) = \prod 1/(1-w^n)$	Base case: $\psi_\pi(\gamma, \emptyset) = (\gamma, \ldots, \gamma)$

5.4 Commutativity of local rules

The algebraic fact that commuting different pairs of operators is independent (when the positions are non-adjacent) has the following bijective counterpart:

Lemma 7 (PhD thesis, Lemma in §3)**.** *If π has inversions at both positions i and j with $|i-j| \geq 2$, then:*

$$\mathfrak{L}_i \circ \mathfrak{L}_j[\psi_\pi] = \mathfrak{L}_j \circ \mathfrak{L}_i[\psi_\pi].$$

This is because $\mathfrak{L}_i$ only affects the $(i+1)$-th diagonal while $\mathfrak{L}_j$ only affects the $(j+1)$-th, so they act on disjoint data. This mirrors the fact that the operators $\Omega[Xu_i]$ and $\Omega[Xu_j]$ commute when they are of the same type.

5.5 Weight preservation

The bijection is not merely weight-preserving but **strongly** weight-preserving: if $\psi_\pi(\gamma, \mathfrak{d}) = (\mu^0, \ldots, \mu^T)$ then for all $1 \leq k \leq T$:

$$|\mu^k| = |\gamma| + |\mathrm{diag}(k)|_{\mathfrak{d}}.$$

The proof proceeds by induction, using the weight condition (**??**) of the local rule and the identity:

$$|\mathrm{diag}(i)|_{\mathfrak{d}'} = m + |\mathrm{diag}(i-1)|_{\mathfrak{d}} + |\mathrm{diag}(i+1)|_{\mathfrak{d}} - |\mathrm{diag}(i)|_{\mathfrak{d}}.$$

6 Summary: the three levels

The relationship between the three works can be visualized as three levels of the same structure:

Level 1: Schur functions (Masters thesis, Part 1).

- Operators: $\mathcal{T}(z)$, $\mathcal{P}(z)$, $C(z)$, $R(z)$ on Λ.
- Fundamental commutation: $\mathcal{T}(v)\mathcal{P}(w) = \Omega(vw)\,\mathcal{P}(w)\mathcal{T}(v)$.
- Consequence: Pieri rules, Jacobi–Trudi, all LR-type bases via U_f.
- Generalization: Stirling numbers, Lah numbers as change-of-basis matrices for LR bases associated to $f(z) = \exp(z) - 1$.

Level 2: Macdonald polynomials (CPP Part I).

- Operators: $\Omega_{q,t}[Xz]$, $\Omega^*_{q,t}[Xz]$, D_z on $\Lambda_{q,t}$.
- Fundamental commutation: $\Omega^*_{q,t}[Xu]\,\Omega_{q,t}[Xv] = \frac{(tuv;q)_\infty}{(quv;q)_\infty}\,\Omega_{q,t}[Xv]\,\Omega^*_{q,t}[Xu]$.
- Consequence: (q,t)-Borodin identity for cylindric plane partitions; at $q = t$ recovers Schur case; at $t = 0$ gives Hall–Littlewood.
- The weight function $W_{\mathfrak{c}}(q,t) = \Omega[(q-t)\mathcal{D}_{\mathfrak{c}}(q,t)]$ has an explicit combinatorial description in terms of peak and valley cubes.

Level 3: Growth diagrams (PhD thesis, Chapter 3).

- Objects: Cylindric growth diagrams on the poset $\mathfrak{G}(n,m) = \mathbb{Z}^2/(x,y) \sim (x+n, y-m)$.
- Local rule $\mathfrak{U}_{\alpha,\beta}$: the bijective incarnation of one commutation step.
- Each application of $\mathfrak{L}_i$ (local rule as higher-order function) $\longleftrightarrow$ one commutation $\Omega^* \leftrightarrow \Omega$.
- Cylinder rotation $\longleftrightarrow$ cyclic invariance of trace $\mathrm{tr}(AB) = \mathrm{tr}(BA)$.
- The full bijection $\psi_\pi : (\mathrm{IP}, \mathrm{ALCD}(\pi)) \to \mathrm{CPP}(\pi)$ is constructed by iterating local rules and rotations, exactly mirroring the algebraic proof.
- Two natural local rules exist (RSK and Burge), related by the Schützenberger involution. Their composition $\psi_R \circ \psi_B^{-1} : \mathrm{CPP}(\pi) \to \mathrm{CPP}(\pi)$ has a conjectural interpretation via crystal bases for $\widehat{\mathfrak{sl}}_n$.

The power of this framework is that the *same* local rule, applied repeatedly on a cylindric growth diagram, produces the full bijective proof of Borodin's product formula. The algebraic proof and the bijective proof are not merely analogous — they are **structurally identical**, with each algebraic manipulation corresponding to a precisely defined combinatorial operation.

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